

Name: Group: Semester: Spring 2026 Time allowed: 15 mins Course: Information Security	Roll No: Examination: Mid Term (Part I) Total marks: 12 Date: April 01, 2026 Instructor: Dr. Tamleek Ali Tanveer
--	---

Total marks for this part are 12. Concentrate on your own paper as your peers may have a different version the paper.

1. Encircle appropriate option(s). [12]

- i. Which key size is commonly used in AES?

a) 64 bits
b) 128 bits
c) 512 bits
d) 1024 bits
- ii. In an access control matrix, rows represent _____.

a) users or subjects
b) objects or resources
c) encryption keys
d) network ports
- iii. A _____ cipher replaces one character with another character.

a) substitution
b) transposition
c) either (a) or (b)
d) neither (a) nor (b)
- iv. In access control, a capability refers to _____.

a) an unforgeable token granting access to a resource
b) a list of users allowed for an object

c) a password used for authentication
d) a log of system activities
- v. Authorization is used to determine _____.

a) whether access should be granted
b) the identity of the user

c) how encryption is applied
d) how passwords are stored
- vi. In an access control matrix, rows represent _____.

a) users or subjects
b) objects or resources

c) encryption keys
d) network ports
- vii. What is the primary purpose of adding a salt to a password before hashing?

a) To reduce password length
b) To make passwords easier to remember

c) To ensure identical passwords produce different hashes
d) To encrypt passwords
- viii. Which one is the correct sequence?

a) Authorization → Authentication → Resource Access
b) Authentication → Authorization → Resource Access
c) Resource Access → Authorization → Authentication
d) Both a and b

What is the purpose of S-boxes in S-DES?

- a) To increase key size
b) To perform substitution and introduce confusion
c) To swap left and right halves
d) To expand the plaintext size
- ix. Before accessing any resource _____
- a) A user must have write access
b) A user must be logged into the operating system
c) A user may not have access right to the resource
d) Both a and b
- x. In AES encryption using CBC mode, what is the primary role of the Initialization Vector (IV)?

- a) To act as the encryption key
 - b) To ensure identical plaintext blocks produce different ciphertexts
 - c) To reduce the size of ciphertext
 - d) To replace the substitution step
- xi. The _____ criterion states that it must be extremely difficult or impossible to create the message if the message digest is given.
- a) strong-collision-resistance
 - b) weak-collision-resistance
 - c) one-wayness
 - d) none of the above
- xii. In CIA the letter 'A' stands for _____
- a) Authentication
 - b) Availability
 - c) Authority
 - d) Access Control
- xiii. What are the properties of a hash function?
- a) One way
 - b) Strong encryption
 - c) Collision resistant
 - d) All of the above
- xiv. Which ONE is considered the best method to distribute Public Keys:
- a) public announcement
 - b) publicly available directory
 - c) public-key authority
 - d) public-key certificates
- xv. In public-key cryptosystem, who know the private key of key owner:
- a) Certification Authority
 - b) Only the owner
 - c) Both CA and Owner
 - d) Receiver and CA
- xvi. Data breaches violate which of the following security requirements?
- a) Integrity
 - b) Availability
 - c) Confidentiality
 - d) Non-repudiation
- xvii. A number of online banking systems send a limited lifetime PIN to your smartphone for you to be able to authenticate yourself to the bank. Is this an example of...
- a) Something you have
 - b) Something you are
 - c) Something you know
 - d) OTP
- xviii. Which security property ensures that a sender cannot deny sending a message?
- a) Confidentiality
 - b) Integrity
 - c) Authentication
 - d) Non-repudiation
- xix. An attacker correctly guesses Alice's password and logs in as her. Is this a case of...
- a) False positive
 - b) True positive
 - c) False negative
 - d) True negative
- xx. A randomly chosen password has six characters that include upper and lower case letters, digits (0-9). In the worst case, how many attempts must a brute-force method make to determine a password when its hashed value is available?
- a) 67^2
 - b) 62^6
 - c) 72^6
 - d) 72^7
- xxi. Brute-force attack is a guaranteed way to break any password. What can be the way to overcome the brute-force attack?
- a) Social Engineering
 - b) Limiting number of password attempts
 - c) Input validation
 - d) None of the above
- xxii. To decrypt using DES, just run the same algorithm with reverse key order.
- a) True
 - b) False
- xxiii. Which of the following is NOT property of a Hash function?
- a) Given message m, It should be easy to compute $H(m)$
 - b) Given m_1 , it is computationally impossible to find m_2 s.t. $H(m_2) = H(m_1)$
 - c) Given $H(m)$, there is no easy way to find m
 - d) Any size of message generates fixed size (128-512) output.
- xxiv. A multi-factor authentication is likely to reduce _____.
- a) True Negative
 - b) False Positive
 - c) False Negative
 - d) Both b and c
- xxv. Reference Monitor _____.
- a) Is part of the TCB
 - b) is not part of the TCB
 - c) is the only part of the TCB
 - d) is the only trusted part of the TCB